DBMS LAB · ASSIGNMENT 5

Database Creation and Table Implementation

Implementing the finalized airline reservation schema in MySQL with SQL DDL

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PROJECT
Airline Reservation & Flight Booking System

DATABASE
MySQL 8.0 · `airline_reservation`

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Aim

To create the `airline_reservation` database and implement the nine tables of the finalized design from Assignment 3 with SQL DDL. Every table gets a primary key and every relationship a foreign key. The structure is then verified from MySQL's data dictionary.

Software Used

Component	Details
Database server	MySQL Community Server 8.0.46, InnoDB storage engine
Client	MySQL 8.0 command-line client
Character set	utf8mb4 / utf8mb4_0900_ai_ci (names in any script, the ₹ symbol)
Script file	`sql/01_create_database.sql`

Schema to Be Implemented

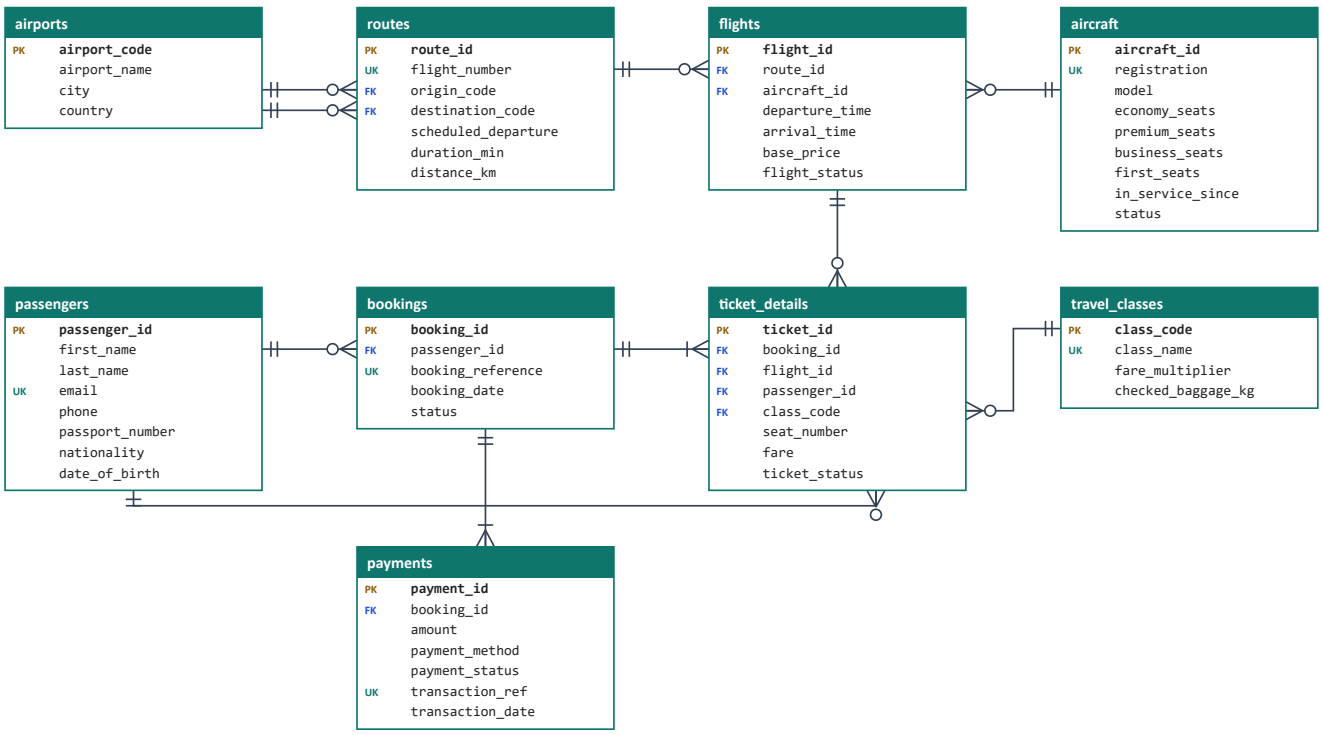


Figure 1. The nine tables and their relationships.

Relationship	Type	Implemented by
airports – routes (origin)	1 : N	routes.origin_code → airports
airports – routes (destination)	1 : N	routes.destination_code → airports
routes – flights	1 : N	flights.route_id → routes
aircraft – flights	1 : N	flights.aircraft_id → aircraft
passengers – bookings	1 : N	bookings.passenger_id → passengers
bookings – ticket_details	1 : N	ticket_details.booking_id → bookings
flights – ticket_details	1 : N	ticket_details.flight_id → flights
passengers – ticket_details (traveller)	1 : N	ticket_details.passenger_id → passengers
travel_classes – ticket_details	1 : N	ticket_details.class_code → travel_classes
bookings – payments	1 : N (one successful)	payments.booking_id → bookings
bookings – flights	M : N	resolved through ticket_details

Step 1 – Create the Database

```
DROP DATABASE IF EXISTS airline_reservation;
CREATE DATABASE airline_reservation
    DEFAULT CHARACTER SET utf8mb4
    DEFAULT COLLATE utf8mb4_0900_ai_ci;
USE airline_reservation;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DROP DATABASE IF EXISTS airline_reservation;
Query OK, 11 rows affected (0.12 sec)

mysql> CREATE DATABASE airline_reservation
    ->     DEFAULT CHARACTER SET utf8mb4
    ->     DEFAULT COLLATE utf8mb4_0900_ai_ci;
Query OK, 1 row affected (0.01 sec)
```

Step 2 – Create the Tables

Parent tables are created before the tables that reference them, and every constraint has an explicit name. Airports use their natural IATA code as the primary key, travel classes use a two-letter code, and all other tables use AUTO_INCREMENT surrogate keys.

Reference tables

These have no foreign keys and are created first.

2.1 airports

Keyed by the three-letter IATA code, a natural key that is short, stable and already used everywhere.

```
CREATE TABLE airports (
    airport_code  CHAR(3)       NOT NULL,
    airport_name  VARCHAR(100)  NOT NULL,
    city          VARCHAR(50)   NOT NULL,
    country       VARCHAR(50)   NOT NULL,
    PRIMARY KEY (airport_code)
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE airports (
    ->     airport_code  CHAR(3)       NOT NULL,
    ->     airport_name  VARCHAR(100)  NOT NULL,
    ->     city          VARCHAR(50)   NOT NULL,
    ->     country       VARCHAR(50)   NOT NULL,
    ->     PRIMARY KEY (airport_code)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

2.2 aircraft

The fleet. Seat counts per class live here, because capacity is a property of the aircraft.

```
CREATE TABLE aircraft (
    aircraft_id       SMALLINT     NOT NULL AUTO_INCREMENT,
    registration      VARCHAR(10)  NOT NULL,
    model             VARCHAR(40)  NOT NULL,
    economy_seats     SMALLINT     NOT NULL,
    premium_seats     SMALLINT     NOT NULL,
    business_seats    SMALLINT     NOT NULL,
    first_seats       SMALLINT     NOT NULL,
    in_service_since  DATE         NOT NULL,
    status            ENUM('Active', 'Maintenance', 'Retired') NOT NULL,
    PRIMARY KEY (aircraft_id)
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE aircraft (
    ->     aircraft_id       SMALLINT     NOT NULL AUTO_INCREMENT,
    ->     registration      VARCHAR(10)  NOT NULL,
    ->     model             VARCHAR(40)  NOT NULL,
    ->     economy_seats     SMALLINT     NOT NULL,
    ->     premium_seats     SMALLINT     NOT NULL,
    ->     business_seats    SMALLINT     NOT NULL,
    ->     first_seats       SMALLINT     NOT NULL,
    ->     in_service_since  DATE         NOT NULL,
    ->     status            ENUM('Active', 'Maintenance', 'Retired') NOT NULL,
    ->     PRIMARY KEY (aircraft_id)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

2.3 travel_classes

Economy, Premium Economy, Business and First, with their fare multipliers and baggage allowances. The two-letter code is the key.

```
CREATE TABLE travel_classes (
    class_code          CHAR(2)       NOT NULL,
    class_name          VARCHAR(20)   NOT NULL,
    fare_multiplier     DECIMAL(4,2)  NOT NULL,
    checked_baggage_kg  TINYINT       NOT NULL,
    PRIMARY KEY (class_code)
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE travel_classes (
    ->     class_code          CHAR(2)       NOT NULL,
    ->     class_name          VARCHAR(20)   NOT NULL,
    ->     fare_multiplier     DECIMAL(4,2)  NOT NULL,
    ->     checked_baggage_kg  TINYINT       NOT NULL,
    ->     PRIMARY KEY (class_code)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

Network and schedule

A route connects two airports; a flight is a dated operation of a route on an aircraft.

2.4 routes

A flight number such as SW101 with its origin, destination, scheduled time, duration and distance. Its two foreign keys reference airports for origin and destination.

```
CREATE TABLE routes (
    route_id             SMALLINT     NOT NULL AUTO_INCREMENT,
    flight_number        VARCHAR(8)   NOT NULL,
    origin_code          CHAR(3)      NOT NULL,
    destination_code     CHAR(3)      NOT NULL,
    scheduled_departure  TIME         NOT NULL,
    duration_min         SMALLINT     NOT NULL,
    distance_km          SMALLINT     NOT NULL,
    PRIMARY KEY (route_id),
    -- IATA codes never change, so plain RESTRICT is used; it also lets
    -- a CHECK constraint compare origin and destination (MySQL rule)
    CONSTRAINT fk_route_origin FOREIGN KEY (origin_code)
        REFERENCES airports (airport_code)
        ON UPDATE RESTRICT ON DELETE RESTRICT,
    CONSTRAINT fk_route_destination FOREIGN KEY (destination_code)
        REFERENCES airports (airport_code)
        ON UPDATE RESTRICT ON DELETE RESTRICT
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE routes (
    ->     route_id             SMALLINT     NOT NULL AUTO_INCREMENT,
    ->     flight_number        VARCHAR(8)   NOT NULL,
    ->     origin_code          CHAR(3)      NOT NULL,
    ->     destination_code     CHAR(3)      NOT NULL,
    ->     scheduled_departure  TIME         NOT NULL,
    ->     duration_min         SMALLINT     NOT NULL,
    ->     distance_km          SMALLINT     NOT NULL,
    ->     PRIMARY KEY (route_id),
    ->
    ->
    ->     CONSTRAINT fk_route_origin FOREIGN KEY (origin_code)
    ->         REFERENCES airports (airport_code)
    ->         ON UPDATE RESTRICT ON DELETE RESTRICT,
    ->     CONSTRAINT fk_route_destination FOREIGN KEY (destination_code)
    ->         REFERENCES airports (airport_code)
    ->         ON UPDATE RESTRICT ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)
```

2.5 flights

One row per dated departure: route, aircraft, actual times, the base price for that day and the operational status.

```sql
CREATE TABLE flights (
    flight_id       INT           NOT NULL AUTO_INCREMENT,
    route_id        SMALLINT      NOT NULL,
    aircraft_id     SMALLINT      NOT NULL,
    departure_time  DATETIME      NOT NULL,
    arrival_time    DATETIME      NOT NULL,
    base_price      DECIMAL(10,2) NOT NULL,
    flight_status   ENUM('Scheduled', 'Delayed', 'Departed', 'Arrived', 'Cancelled') NOT NULL,
    PRIMARY KEY (flight_id),
    CONSTRAINT fk_flight_route FOREIGN KEY (route_id)
        REFERENCES routes (route_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_flight_aircraft FOREIGN KEY (aircraft_id)
        REFERENCES aircraft (aircraft_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE flights (
    ->     flight_id       INT           NOT NULL AUTO_INCREMENT,
    ->     route_id        SMALLINT      NOT NULL,
    ->     aircraft_id     SMALLINT      NOT NULL,
    ->     departure_time  DATETIME      NOT NULL,
    ->     arrival_time    DATETIME      NOT NULL,
    ->     base_price      DECIMAL(10,2) NOT NULL,
    ->     flight_status   ENUM('Scheduled', 'Delayed', 'Departed', 'Arrived', 'Cancelled') NOT NULL,
    ->     PRIMARY KEY (flight_id),
    ->     CONSTRAINT fk_flight_route FOREIGN KEY (route_id)
    ->         REFERENCES routes (route_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_flight_aircraft FOREIGN KEY (aircraft_id)
    ->         REFERENCES aircraft (aircraft_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.07 sec)
```

Customers and bookings

A passenger makes bookings; each ticket places one traveller on one flight in one class; payments record every attempt to pay for a booking.

2.6 passengers

Customer and traveller profiles, including the passport details needed at check-in.

```sql
CREATE TABLE passengers (
    passenger_id     INT           NOT NULL AUTO_INCREMENT,
    first_name       VARCHAR(50)   NOT NULL,
    last_name        VARCHAR(50)   NOT NULL,
    email            VARCHAR(100)  NOT NULL,
    phone            VARCHAR(15)   NOT NULL,
    passport_number  VARCHAR(20)   NOT NULL,
    nationality      VARCHAR(50)   NOT NULL,
    date_of_birth    DATE          NOT NULL,
    PRIMARY KEY (passenger_id)
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE passengers (
    ->     passenger_id     INT           NOT NULL AUTO_INCREMENT,
    ->     first_name       VARCHAR(50)   NOT NULL,
    ->     last_name        VARCHAR(50)   NOT NULL,
    ->     email            VARCHAR(100)  NOT NULL,
    ->     phone            VARCHAR(15)   NOT NULL,
    ->     passport_number  VARCHAR(20)   NOT NULL,
    ->     nationality      VARCHAR(50)   NOT NULL,
    ->     date_of_birth    DATE          NOT NULL,
    ->     PRIMARY KEY (passenger_id)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

2.7 bookings

One reservation (PNR). `passenger_id` is the person who made the booking.

```
CREATE TABLE bookings (
    booking_id         INT          NOT NULL AUTO_INCREMENT,
    passenger_id       INT          NOT NULL,          -- who made the booking
    booking_reference  CHAR(6)      NOT NULL,          -- the PNR
    booking_date       DATETIME     NOT NULL,
    status             ENUM('Pending', 'Confirmed', 'Cancelled') NOT NULL,
    PRIMARY KEY (booking_id),
    CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id)
        REFERENCES passengers (passenger_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE bookings (
    ->     booking_id         INT          NOT NULL AUTO_INCREMENT,
    ->     passenger_id       INT          NOT NULL,
    ->     booking_reference  CHAR(6)      NOT NULL,
    ->     booking_date       DATETIME     NOT NULL,
    ->     status             ENUM('Pending', 'Confirmed', 'Cancelled') NOT NULL,
    ->     PRIMARY KEY (booking_id),
    ->     CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id)
    ->         REFERENCES passengers (passenger_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

2.8 ticket_details

One seat on one flight for one traveller. It references four parents: booking, flight, traveller and travel class.

```
CREATE TABLE ticket_details (
    ticket_id      INT            NOT NULL AUTO_INCREMENT,
    booking_id     INT            NOT NULL,
    flight_id      INT            NOT NULL,
    passenger_id   INT            NOT NULL,            -- who is flying
    class_code     CHAR(2)        NOT NULL,
    seat_number    VARCHAR(4)     NOT NULL,
    fare           DECIMAL(10,2)  NOT NULL,            -- price charged (snapshot)
    ticket_status  ENUM('Booked', 'Checked-in', 'Boarded', 'No-show', 'Cancelled') NOT NULL,
    PRIMARY KEY (ticket_id),
    CONSTRAINT fk_ticket_booking FOREIGN KEY (booking_id)
        REFERENCES bookings (booking_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_ticket_flight FOREIGN KEY (flight_id)
        REFERENCES flights (flight_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_ticket_passenger FOREIGN KEY (passenger_id)
        REFERENCES passengers (passenger_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_ticket_class FOREIGN KEY (class_code)
        REFERENCES travel_classes (class_code)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE ticket_details (
    ->     ticket_id      INT            NOT NULL AUTO_INCREMENT,
    ->     booking_id     INT            NOT NULL,
    ->     flight_id      INT            NOT NULL,
    ->     passenger_id   INT            NOT NULL,
    ->     class_code     CHAR(2)        NOT NULL,
    ->     seat_number    VARCHAR(4)     NOT NULL,
    ->     fare           DECIMAL(10,2)  NOT NULL,
    ->     ticket_status  ENUM('Booked', 'Checked-in', 'Boarded', 'No-show', 'Cancelled') NOT NULL,
    ->     PRIMARY KEY (ticket_id),
    ->     CONSTRAINT fk_ticket_booking FOREIGN KEY (booking_id)
    ->         REFERENCES bookings (booking_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_ticket_flight FOREIGN KEY (flight_id)
    ->         REFERENCES flights (flight_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_ticket_passenger FOREIGN KEY (passenger_id)
    ->         REFERENCES passengers (passenger_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_ticket_class FOREIGN KEY (class_code)
    ->         REFERENCES travel_classes (class_code)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)
```

2.9 payments

Every payment attempt (failed, pending, successful or refunded) with its amount and gateway transaction reference.

```
CREATE TABLE payments (
    payment_id        INT            NOT NULL AUTO_INCREMENT,
    booking_id        INT            NOT NULL,
    amount            DECIMAL(10,2)  NOT NULL,
    payment_method    ENUM('Credit Card', 'Debit Card', 'Net Banking', 'UPI') NOT NULL,
    payment_status    ENUM('Pending', 'Success', 'Failed', 'Refunded') NOT NULL,
    transaction_ref   VARCHAR(20)    NOT NULL,
    transaction_date  DATETIME       NOT NULL,
    PRIMARY KEY (payment_id),
    CONSTRAINT fk_payment_booking FOREIGN KEY (booking_id)
        REFERENCES bookings (booking_id)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE = InnoDB;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE TABLE payments (
    ->     payment_id        INT            NOT NULL AUTO_INCREMENT,
    ->     booking_id        INT            NOT NULL,
    ->     amount            DECIMAL(10,2)  NOT NULL,
    ->     payment_method    ENUM('Credit Card', 'Debit Card', 'Net Banking', 'UPI') NOT NULL,
    ->     payment_status    ENUM('Pending', 'Success', 'Failed', 'Refunded') NOT NULL,
    ->     transaction_ref   VARCHAR(20)    NOT NULL,
    ->     transaction_date  DATETIME       NOT NULL,
    ->     PRIMARY KEY (payment_id),
    ->     CONSTRAINT fk_payment_booking FOREIGN KEY (booking_id)
    ->         REFERENCES bookings (booking_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)
```

Step 3 – Views for Derived Information

The booking total and a flight's seats sold are derived values (Assignment 3, item 4), so they are computed by views instead of being stored.

```
CREATE VIEW v_booking_totals AS
SELECT b.booking_id, b.booking_reference, b.status,
       COUNT(t.ticket_id)                                        AS active_tickets,
       COALESCE(SUM(t.fare), 0)                                  AS total_fare,
       COALESCE((SELECT SUM(p.amount) FROM payments p
                 WHERE p.booking_id = b.booking_id
                   AND p.payment_status = 'Success'), 0)         AS amount_paid
FROM bookings b
LEFT JOIN ticket_details t
       ON t.booking_id = b.booking_id AND t.ticket_status <> 'Cancelled'
GROUP BY b.booking_id, b.booking_reference, b.status;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE VIEW v_booking_totals AS
    -> SELECT b.booking_id, b.booking_reference, b.status,
    ->        COUNT(t.ticket_id)                                        AS active_tickets,
    ->        COALESCE(SUM(t.fare), 0)                                  AS total_fare,
    ->        COALESCE((SELECT SUM(p.amount) FROM payments p
    ->                  WHERE p.booking_id = b.booking_id
    ->                    AND p.payment_status = 'Success'), 0)         AS amount_paid
    -> FROM bookings b
    -> LEFT JOIN ticket_details t
    ->        ON t.booking_id = b.booking_id AND t.ticket_status <> 'Cancelled'
    -> GROUP BY b.booking_id, b.booking_reference, b.status;
Query OK, 0 rows affected (0.01 sec)
```

```sql
CREATE VIEW v_flight_occupancy AS
SELECT f.flight_id, r.flight_number, r.origin_code, r.destination_code,
       f.departure_time, f.flight_status,
       a.economy_seats + a.premium_seats + a.business_seats + a.first_seats AS capacity,
       COUNT(t.ticket_id)                                                    AS seats_sold,
       ROUND(100 * COUNT(t.ticket_id)
             / (a.economy_seats + a.premium_seats + a.business_seats + a.first_seats), 1)
                                                                             AS load_factor_pct
FROM flights f
JOIN routes r   ON r.route_id    = f.route_id
JOIN aircraft a ON a.aircraft_id = f.aircraft_id
LEFT JOIN ticket_details t
       ON t.flight_id = f.flight_id AND t.ticket_status <> 'Cancelled'
GROUP BY f.flight_id, r.flight_number, r.origin_code, r.destination_code,
         f.departure_time, f.flight_status,
         a.economy_seats, a.premium_seats, a.business_seats, a.first_seats;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> CREATE VIEW v_flight_occupancy AS
    -> SELECT f.flight_id, r.flight_number, r.origin_code, r.destination_code,
    ->        f.departure_time, f.flight_status,
    ->        a.economy_seats + a.premium_seats + a.business_seats + a.first_seats AS capacity,
    ->        COUNT(t.ticket_id)                                                    AS seats_sold,
    ->        ROUND(100 * COUNT(t.ticket_id)
    ->              / (a.economy_seats + a.premium_seats + a.business_seats + a.first_seats), 1)
    ->                                                                              AS load_factor_pct
    -> FROM flights f
    -> JOIN routes r   ON r.route_id    = f.route_id
    -> JOIN aircraft a ON a.aircraft_id = f.aircraft_id
    -> LEFT JOIN ticket_details t
    ->        ON t.flight_id = f.flight_id AND t.ticket_status <> 'Cancelled'
    -> GROUP BY f.flight_id, r.flight_number, r.origin_code, r.destination_code,
    ->          f.departure_time, f.flight_status,
    ->          a.economy_seats, a.premium_seats, a.business_seats, a.first_seats;
Query OK, 0 rows affected (0.01 sec)
```

Step 4 – Verification

4.1 Objects created

```sql
SHOW FULL TABLES FROM airline_reservation;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> SHOW FULL TABLES FROM airline_reservation;
+-------------------------------+------------+
| Tables_in_airline_reservation | Table_type |
+-------------------------------+------------+
| aircraft                      | BASE TABLE |
| airports                      | BASE TABLE |
| bookings                      | BASE TABLE |
| flights                       | BASE TABLE |
| passengers                    | BASE TABLE |
| payments                      | BASE TABLE |
| routes                        | BASE TABLE |
| ticket_details                | BASE TABLE |
| travel_classes                | BASE TABLE |
| v_booking_totals              | VIEW       |
| v_flight_occupancy            | VIEW       |
+-------------------------------+------------+
11 rows in set (0.00 sec)
```

4.2 Table structures

```
DESCRIBE airports;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE airports;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| airport_code | char(3)      | NO   | PRI | NULL    |       |
| airport_name | varchar(100) | NO   |     | NULL    |       |
| city         | varchar(50)  | NO   |     | NULL    |       |
| country      | varchar(50)  | NO   |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

```
DESCRIBE aircraft;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE aircraft;
+------------------+---------------------------------------+------+-----+---------+----------------+
| Field            | Type                                  | Null | Key | Default | Extra          |
+------------------+---------------------------------------+------+-----+---------+----------------+
| aircraft_id      | smallint                              | NO   | PRI | NULL    | auto_increment |
| registration     | varchar(10)                           | NO   |     | NULL    |                |
| model            | varchar(40)                           | NO   |     | NULL    |                |
| economy_seats    | smallint                              | NO   |     | NULL    |                |
| premium_seats    | smallint                              | NO   |     | NULL    |                |
| business_seats   | smallint                              | NO   |     | NULL    |                |
| first_seats      | smallint                              | NO   |     | NULL    |                |
| in_service_since | date                                  | NO   |     | NULL    |                |
| status           | enum('Active','Maintenance','Retired') | NO   |     | NULL    |                |
+------------------+---------------------------------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
DESCRIBE travel_classes;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE travel_classes;
+--------------------+--------------+------+-----+---------+-------+
| Field              | Type         | Null | Key | Default | Extra |
+--------------------+--------------+------+-----+---------+-------+
| class_code         | char(2)      | NO   | PRI | NULL    |       |
| class_name         | varchar(20)  | NO   |     | NULL    |       |
| fare_multiplier    | decimal(4,2) | NO   |     | NULL    |       |
| checked_baggage_kg | tinyint      | NO   |     | NULL    |       |
+--------------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

```
DESCRIBE routes;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE routes;
+---------------------+------------+------+-----+---------+----------------+
| Field               | Type       | Null | Key | Default | Extra          |
+---------------------+------------+------+-----+---------+----------------+
| route_id            | smallint   | NO   | PRI | NULL    | auto_increment |
| flight_number       | varchar(8) | NO   |     | NULL    |                |
| origin_code         | char(3)    | NO   | MUL | NULL    |                |
| destination_code    | char(3)    | NO   | MUL | NULL    |                |
| scheduled_departure | time       | NO   |     | NULL    |                |
| duration_min        | smallint   | NO   |     | NULL    |                |
| distance_km         | smallint   | NO   |     | NULL    |                |
+---------------------+------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
DESCRIBE flights;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE flights;
+---------------+-----------------------------------------------------------------+------+-----+---------+----------------+
| Field         | Type                                                            | Null | Key | Default | Extra          |
+---------------+-----------------------------------------------------------------+------+-----+---------+----------------+
| flight_id     | int                                                             | NO   | PRI | NULL    | auto_increment |
| route_id      | smallint                                                        | NO   | MUL | NULL    |                |
| aircraft_id   | smallint                                                        | NO   | MUL | NULL    |                |
| departure_time | datetime                                                       | NO   |     | NULL    |                |
| arrival_time  | datetime                                                        | NO   |     | NULL    |                |
| base_price    | decimal(10,2)                                                   | NO   |     | NULL    |                |
| flight_status | enum('Scheduled','Delayed','Departed','Arrived','Cancelled') | NO   |     | NULL    |                |
+---------------+-----------------------------------------------------------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
DESCRIBE passengers;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE passengers;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| passenger_id    | int          | NO   | PRI | NULL    | auto_increment |
| first_name      | varchar(50)  | NO   |     | NULL    |                |
| last_name       | varchar(50)  | NO   |     | NULL    |                |
| email           | varchar(100) | NO   |     | NULL    |                |
| phone           | varchar(15)  | NO   |     | NULL    |                |
| passport_number | varchar(20)  | NO   |     | NULL    |                |
| nationality     | varchar(50)  | NO   |     | NULL    |                |
| date_of_birth   | date         | NO   |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
DESCRIBE bookings;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE bookings;
+-------------------+-----------------------------------------+------+-----+---------+----------------+
| Field             | Type                                    | Null | Key | Default | Extra          |
+-------------------+-----------------------------------------+------+-----+---------+----------------+
| booking_id        | int                                     | NO   | PRI | NULL    | auto_increment |
| passenger_id      | int                                     | NO   | MUL | NULL    |                |
| booking_reference | char(6)                                 | NO   |     | NULL    |                |
| booking_date      | datetime                                | NO   |     | NULL    |                |
| status            | enum('Pending','Confirmed','Cancelled') | NO   |     | NULL    |                |
+-------------------+-----------------------------------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
DESCRIBE ticket_details;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE ticket_details;
+--------------+-----------------------------------------------------------+------+-----+---------+----------------+
| Field        | Type                                                      | Null | Key | Default | Extra          |
+--------------+-----------------------------------------------------------+------+-----+---------+----------------+
| ticket_id    | int                                                       | NO   | PRI | NULL    | auto_increment |
| booking_id   | int                                                       | NO   | MUL | NULL    |                |
| flight_id    | int                                                       | NO   | MUL | NULL    |                |
| passenger_id | int                                                       | NO   | MUL | NULL    |                |
| class_code   | char(2)                                                   | NO   | MUL | NULL    |                |
| seat_number  | varchar(4)                                                | NO   |     | NULL    |                |
| fare         | decimal(10,2)                                             | NO   |     | NULL    |                |
| ticket_status | enum('Booked','Checked-in','Boarded','No-show','Cancelled') | NO   |     | NULL    |                |
+--------------+-----------------------------------------------------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
DESCRIBE payments;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DESCRIBE payments;
+-----------------+-----------------------------------------------------+------+-----+---------+----------------+
| Field           | Type                                                | Null | Key | Default | Extra          |
+-----------------+-----------------------------------------------------+------+-----+---------+----------------+
| payment_id      | int                                                 | NO   | PRI | NULL    | auto_increment |
| booking_id      | int                                                 | NO   | MUL | NULL    |                |
| amount          | decimal(10,2)                                       | NO   |     | NULL    |                |
| payment_method  | enum('Credit Card','Debit Card','Net Banking','UPI') | NO   |     | NULL    |                |
| payment_status  | enum('Pending','Success','Failed','Refunded')       | NO   |     | NULL    |                |
| transaction_ref | varchar(20)                                         | NO   |     | NULL    |                |
| transaction_date | datetime                                           | NO   |     | NULL    |                |
+-----------------+-----------------------------------------------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

4.3 Primary keys

```
SELECT TABLE_NAME, GROUP_CONCAT(COLUMN_NAME) AS primary_key
FROM information_schema.KEY_COLUMN_USAGE
WHERE TABLE_SCHEMA = 'airline_reservation' AND CONSTRAINT_NAME = 'PRIMARY'
GROUP BY TABLE_NAME;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> SELECT TABLE_NAME, GROUP_CONCAT(COLUMN_NAME) AS primary_key
    -> FROM information_schema.KEY_COLUMN_USAGE
    -> WHERE TABLE_SCHEMA = 'airline_reservation' AND CONSTRAINT_NAME = 'PRIMARY'
    -> GROUP BY TABLE_NAME;
+----------------+--------------+
| TABLE_NAME     | primary_key  |
+----------------+--------------+
| aircraft       | aircraft_id  |
| airports       | airport_code |
| bookings       | booking_id   |
| flights        | flight_id    |
| passengers     | passenger_id |
| payments       | payment_id   |
| routes         | route_id     |
| ticket_details | ticket_id    |
| travel_classes | class_code   |
+----------------+--------------+
9 rows in set (0.00 sec)
```

4.4 Foreign keys and referential actions

```sql
SELECT rc.TABLE_NAME AS child, kcu.COLUMN_NAME AS fk_column,
       rc.REFERENCED_TABLE_NAME AS parent, kcu.REFERENCED_COLUMN_NAME AS parent_key,
       rc.DELETE_RULE AS on_delete, rc.UPDATE_RULE AS on_update
FROM information_schema.REFERENTIAL_CONSTRAINTS rc
JOIN information_schema.KEY_COLUMN_USAGE kcu
  ON kcu.CONSTRAINT_SCHEMA = rc.CONSTRAINT_SCHEMA
 AND kcu.CONSTRAINT_NAME   = rc.CONSTRAINT_NAME
WHERE rc.CONSTRAINT_SCHEMA = 'airline_reservation'
ORDER BY child, fk_column;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> SELECT rc.TABLE_NAME AS child, kcu.COLUMN_NAME AS fk_column,
    ->        rc.REFERENCED_TABLE_NAME AS parent, kcu.REFERENCED_COLUMN_NAME AS parent_key,
    ->        rc.DELETE_RULE AS on_delete, rc.UPDATE_RULE AS on_update
    -> FROM information_schema.REFERENTIAL_CONSTRAINTS rc
    -> JOIN information_schema.KEY_COLUMN_USAGE kcu
    ->   ON kcu.CONSTRAINT_SCHEMA = rc.CONSTRAINT_SCHEMA
    ->  AND kcu.CONSTRAINT_NAME   = rc.CONSTRAINT_NAME
    -> WHERE rc.CONSTRAINT_SCHEMA = 'airline_reservation'
    -> ORDER BY child, fk_column;
+----------------+------------------+----------------+--------------+-----------+-----------+
| child          | fk_column        | parent         | parent_key   | on_delete | on_update |
+----------------+------------------+----------------+--------------+-----------+-----------+
| bookings       | passenger_id     | passengers     | passenger_id | RESTRICT  | CASCADE   |
| flights        | aircraft_id      | aircraft       | aircraft_id  | RESTRICT  | CASCADE   |
| flights        | route_id         | routes         | route_id     | RESTRICT  | CASCADE   |
| payments       | booking_id       | bookings       | booking_id   | CASCADE   | CASCADE   |
| routes         | destination_code | airports       | airport_code | RESTRICT  | RESTRICT  |
| routes         | origin_code      | airports       | airport_code | RESTRICT  | RESTRICT  |
| ticket_details | booking_id       | bookings       | booking_id   | CASCADE   | CASCADE   |
| ticket_details | class_code       | travel_classes | class_code   | RESTRICT  | CASCADE   |
| ticket_details | flight_id        | flights        | flight_id    | RESTRICT  | CASCADE   |
| ticket_details | passenger_id     | passengers     | passenger_id | RESTRICT  | CASCADE   |
+----------------+------------------+----------------+--------------+-----------+-----------+
10 rows in set (0.05 sec)
```

4.5 Choice of referential actions

Rule	Used on	Reason
RESTRICT	airports → routes, routes / aircraft → flights, flights / passengers / classes → tickets, passengers → bookings	Operational history must survive: an airport with routes, a flight with tickets or a passenger with bookings cannot be deleted. Aircraft are retired by status instead.
CASCADE (delete)	bookings → ticket_details, bookings → payments	Tickets and payment attempts are part of their booking.
CASCADE (update)	all foreign keys except the two airport links	If a key value is ever corrected, the change flows to the child rows.
RESTRICT (update)	routes.origin_code, routes.destination_code	IATA codes never change. MySQL also forbids CHECK constraints on columns used by a cascading foreign key, and the CHECK origin ≠ destination needs these columns (Assignment 6).

Result

The `airline_reservation` database was created with 9 InnoDB tables, 10 foreign keys and 2 views, matching the finalized design. The data dictionary confirms the primary key of every table, a foreign key for every relationship in the ER diagram, and the intended referential action on each. The many-to-many link between bookings and flights is resolved through `ticket_details`.